

WORK AND ENERGY: FURTHER PROBLEMS**20/01/26**

1. A car of mass 1 tonne travelling at 70 mph performs an emergency stop, and the braking distance is 75 m. Find the braking force acting on the car (assumed constant throughout the motion).
2. A particle of mass 3 kg passes a point A travelling at 2.5 ms^{-1} . It is acted on by a driving force of 6 N and a resistance force. It later passes a point B, where $|AB| = 7 \text{ m}$, travelling at $v \text{ ms}^{-1}$. You are given that the work done against resistance force is 25 J.
 - a. Explain why $\left(\frac{1}{2} \times 3 \times 2.5^2\right) + 6 \times 7 - 25 = \left(\frac{1}{2} \times 3 \times v^2\right)$, justifying your choice of sign for the work done terms.
 - b. Solve the equation to find v .
3. A body of mass 10 kg is initially travelling at 2 ms^{-1} along a rough horizontal surface. It moves a distance of 5 m under the influences of a constant driving force of 20 N as well as a frictional force. If the work done against friction while it travels these 5 m is 70 J, find the velocity of the body after it has moved 5 m.
4. A particle of mass 4 kg is acted on by a constant driving force of magnitude 13 N, as well as resistance forces of magnitude $F \text{ N}$. While travelling a distance of 6 m, its velocity decreases from 5 ms^{-1} to 1.5 ms^{-1} . Find F .
5. A bullet of mass 15 grams is fired towards a fixed wooden block, and enters the block when travelling horizontally at 400 ms^{-1} . It comes to rest after penetrating a distance of 25 cm. Find the work done against the resistance of the wood, and find the magnitude of the resistance (assumed constant throughout).
6. A bullet of mass 8 g is fired towards a fixed wooden block and enters the block when travelling horizontally at 300 ms^{-1} . Find, using an energy method, how far the bullet penetrates if the wood provides a constant resistance of 1800 N.
7. A car of mass 1100 kg is travelling at a speed $u \text{ ms}^{-1}$ when it enters a flooded section of road 25 m long. The driver floors it, causing a driving force of 2700 N. A resistance force of $F \text{ N}$ opposes the motion of the car, and it emerges from the flooded section with a speed of $v \text{ ms}^{-1}$.
 - a. If $F = 2750$ and $v = 3$, what is the value of u ?
 - b. If $F = 3500$ and $u = 6$, what happens?
 - c. If $F = 2900$, what is the least possible value of u if the car is to make it through?
 - d. If $u = 12$, what is the greatest possible value of F if the car is to make it through?
8. A tractor of mass 2500 kg is ploughing a straight furrow in a rough field. The driving force on the tractor is 1800 N while there are resistances to motion of 1750 N. The tractor passes a telegraph pole at 0.5 ms^{-1} and leaves the field via a gate at a speed of 2.5 ms^{-1} . Use an energy method to find the distance between the telegraph pole and the gate.
9. A curling stone of mass 18 kg is cast along an ice rink. It leaves the curler's hand at a speed of 4 ms^{-1} and strikes a stationary opposing stone after sliding a distance of 42 m. The captain estimates that the impact must occur at a speed of 0.8 ms^{-1} to leave her stone in the best possible position. The motion of the stone can be modified by two sweepers who may reduce the sliding friction by sweeping the ice in front of the moving stone. The friction is normally 5 N, but when the sweepers sweep, it reduces to 2 N. For how much of the 42 m should the sweepers sweep?